

Predictive Horizons of High-Dimensional Biological Systems: The Connectome Is the Measurement, Not the System

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April 8, 2026

Abstract

Biological systems maintain coherent, nonergodic states against a high-dimensional environment that continuously drives them toward thermal dissolution. We argue that this maintenance—not computation, not representation—is the physical substrate of intelligence. The environment presents many oscillatory modes spanning timescales from milliseconds to hours; the organism must internalize enough of these modes to predict threats and opportunities, but each mode tracked costs metabolic energy. Intelligence is the bandwidth of environmental modes an organism can embody while remaining coherent.

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We test this framework on the complete *C. elegans* connectome (448 neurons, 4,681 synapses, 2,698 gap junctions) using three experimentally derived coupling layers: the synaptic connectome, ephaptic (bioelectric field) coupling from 3D neuron positions, and the neuropeptide network (31,479 extrasynaptic interactions from Ripoll-Sánchez et al. 2023). The environment is a multi-mode oscillatory field with logarithmically spaced timescales and $1/f$ amplitude scaling. Each coupling layer accesses a different band of the environmental spectrum: the connectome routes fast signals, the ephaptic field coordinates medium-timescale spatial responses, and the neuropeptide network tracks slow regime changes. Under metabolic constraint, the connectome alone enters progressive complexity loss—effective dimensionality D_{eff} declines monotonically as the system cannot track the slow modes that signal environmental regime changes. The full trilayer architecture maintains a stable D_{eff} trajectory, finding the optimal operating dimensionality that balances spectral coverage against metabolic cost.

We construct a Landau-Ginzburg coherence field $\varphi(x, t)$ on the worm’s body axis in which coherent states occupy a lower free-energy basin than decoherent states. The coherence domain is low-dimensional at macroscale but high-dimensional at microscale, with low-dimensional defence mechanisms (potential landscape, homeostatic attractors) protecting the high-dimensional interior against noise. Progressive complexity loss—the hallmark of biological aging—corresponds to contraction of the coherent domain as metabolic capacity declines. Low-dimensional observers systematically underestimate the information high-dimensional systems contain, and discrete coding destroys $\sim 85\%$ of predictive content. The connectome is one such low-dimensional observer: it captures synaptic specificity but misses the continuous field dynamics that defend nonergodicity. This is why the OpenWorm project’s connectome-based simulations cannot reproduce the worm’s adaptive behavior: a wiring diagram is a measurement of the system, not the system itself.

Keywords: nonergodic defence, coherence field, connectome, Landau-Ginzburg, effective dimensionality, spectral bandwidth, metabolic constraint, *C. elegans*, complexity loss

On “dimensionality.” Three distinct quantities must not be conflated. (i) The *physical degrees of freedom* N_{phys} : the full state space of the continuous system (fields, phases, concentrations)—effectively unbounded. (ii) The *effective dynamical dimensionality* D_{eff} : how many modes move together coherently (participation ratio, effective rank of covariance spectra). This is the quantity we measure and model. Crucially, D_{eff} *decreases* with coherence: a more organized system concentrates its dynamics onto fewer, more structured modes. (iii) The *observed dimensionality* D_{obs} : what a measurement instrument resolves—an electrode array, a connectome trace, a PCA decomposition. The relationship $N_{\text{phys}} \gg D_{\text{eff}} \geq D_{\text{obs}}$ holds generically. Concentration of measure guarantees that projections from N_{phys} to D_{obs} look smooth and low-dimensional regardless of the system’s true complexity. A low D_{obs} does not imply a simple system—it implies a lossy measurement of a potentially vast state space. A low D_{eff} does not imply a simple system either—it implies a *coherent* one.

1 Introduction

The OpenWorm project set out to simulate *Caenorhabditis elegans* from its complete connectome [1]. The wiring diagram was known: 302 neurons, 6,393 chemical synapses, 890 gap junctions, mapped at electron-microscope resolution by White et al. [2] and refined by Cook et al. [3]. The assumption was natural: given the complete circuit, the behavior should follow. It did not. Despite years of engineering effort, connectome-based simulations of *C. elegans* have not produced the coordinated sinusoidal locomotion that the living worm generates from the moment it hatches [8, 9].

We argue that this failure reveals a category error in computational neuroscience: confusing the measurement for the system. The connectome is what you see when you slice a worm, put it under an electron microscope, and trace connections. It is a low-dimensional projection—a graph—of a continuous, high-dimensional, spatiotemporal biological system.

The graph captures which neurons are connected. It does not capture:

- The continuous bioelectric field in which every neuron is embedded, through which spatially proximate neurons influence each other without synaptic connections (ephaptic coupling [10, 11]);
- The slow, diffuse neuropeptide signals that modulate circuit behavior on timescales of seconds to minutes, broadcast through the extracellular space rather than through specific synaptic connections [12];
- The proprioceptive feedback loop through which the worm senses its own body configuration and uses that information to coordinate the next motor command [13];
- The temporal microstructure of when signals arrive relative to each other—phase relationships that carry information but are invisible to a static wiring diagram.

These are not exotic or speculative channels. They are measured, documented, and well-characterized in the *C. elegans* literature. They are systematically absent from connectome-based simulations because the modeling toolchain—NeuroML, NEURON, Brian2, OpenWorm/c302—is built around the assumption that neurons are the computational units and synapses are the connections. The toolchain cannot represent a continuous field.

The same pattern played out in the history of glial cells: for decades, glia were “just structural support” because the electrophysiological toolchain could not see what they were doing. Now we know they modulate synaptic transmission, maintain ion homeostasis, and form their own signaling networks. They were invisible to the measurement paradigm, not absent from the biology.

The resource is right there. Every neuron in *C. elegans* sits in a bioelectric field that provides continuous, high-bandwidth, spatially graded coupling. Evolution would not leave this resource unused—it is the radio antenna that the graph-based model ignores. This paper demonstrates that the missing coupling channels are what the organism needs not just for motor coordination, but for something more fundamental: defending its nonergodic state

against a high-dimensional environment.

The environment is not a single signal. It is a high-dimensional oscillatory system with modes spanning timescales from milliseconds (vibration, touch) to hours and days (thermal drift, food availability, circadian rhythms). A worm that tracks only fast modes—the band accessible through the connectome’s synaptic channels—misses the slow environmental changes that determine whether it should be foraging, escaping, or entering dauer. The more environmental modes an organism can internalize, the better it can predict and respond. But each additional mode requires metabolic energy to maintain the coupling. Intelligence, in this framework, is the bandwidth of environmental modes the organism can embody while remaining coherent—the balance point between tracking too little (fragile, surprised by slow changes) and tracking too much (metabolically unsustainable, tracking noise).

This framework integrates four prior contributions to this journal. *Limits of Falsifiability* [15] established that high-dimensional systems impose measurement limits: below certain thresholds, internal structure is epistemically inaccessible to low-dimensional observers. *Timing Inaccessibility* [16] showed that irreversible state registration has a thermodynamic floor: at least $k_B T \ln 2$ per bit, imposing a minimum τ_{meas} . *Intelligence as High-Dimensional Coherence* [17] argued that systems with $D_{\text{eff}} \gg 1$ carry rich correlational structure across coupled subsystems, invisible to sparse readout channels. *Coherence Time* [18] derived that oscillatory coordination has its own physical timescale τ_{sync} , set by coupling strength, topology, and coordination depth.

The present paper puts these pieces together on a real biological system. We show that the three coupling layers of *C. elegans*—connectome, ephaptic field, and neuropeptide network—each access a different band of the environmental frequency spectrum. The connectome routes fast, specific signals. The ephaptic field integrates spatially to coordinate medium-timescale responses. The neuropeptide network, because it is slow and diffuse, is the only layer that tracks the slow environmental modes determining behavioral regime. Under metabolic constraint, the connectome alone enters progressive complexity loss: D_{eff} declines

monotonically as the system cannot anticipate slow regime changes. The full trilayer architecture maintains a stable D_{eff} trajectory—not by maximizing dimensionality, but by finding the operating point that balances spectral coverage against metabolic cost. This is what the OpenWorm simulation misses: not computation, but the multi-layer coupling architecture that defends nonergodicity across the full environmental spectrum.

The remainder of the paper is organized as follows. Section 2 develops the sync-measurement asymmetry on thermodynamic and dynamical grounds. Section 3 presents a three-level synchronization hierarchy from passive to selective. Section 4 tests the framework on the real *C. elegans* connectome with a multi-mode environment and metabolic budget. Section 5 demonstrates the observational gap between system information and observer information. Section 6 reframes predictive coding as a description of the low-dimensional shadow, not the mechanism. Section 7 discusses implications for aging, substrate dependence, and the thermodynamics of intelligence.

2 The Sync-Measurement Asymmetry

The central claim rests on an asymmetry between two processes: synchronization (continuous entrainment of internal dynamics to environmental structure) and measurement (irreversible commitment of internal state to a discrete output).

2.1 Why synchronization outruns measurement

Synchronization and measurement differ in channel capacity. Entrainment exploits the *full coupling interface* between organism and environment in parallel. Every receptor, every ion channel, every membrane surface element simultaneously mediates the flow of correlations from external dynamics into internal state. A bacterium swimming in a chemical gradient does not entrain one receptor at a time; the entire array participates. The bandwidth of synchronization scales with the dimensionality of the coupling interface. Recent work on

cytoelectric coupling [10] suggests the interface may be broader than synaptic connectivity alone—endogenous electric fields provide an additional high-bandwidth channel through which neural ensembles entrain to structured input.

Measurement, by contrast, is a bottleneck. To register environmental information as a discrete, irreversible output—a tumble decision, a motor plan, a categorical judgment—the system must stabilize a continuous state, collapse it through a decision boundary, and commit the result. This is the thermodynamically irreversible “commit event” of [18], dissipating at least $k_B T \ln 2$ per bit [16]. The output channel is necessarily low-dimensional (a flagellar motor has one rotational degree of freedom; a saccade has two).

Moreover, the effective dimensionality of a coupling channel exceeds its apparent channel count through temporal precision. Phase coding—encoding information in the temporal placement of signals relative to an ongoing oscillation—multiplies the effective bandwidth of any coupling interface. The temporal microstructure that is thermodynamically inaccessible to discrete measurement remains accessible to continuous entrainment, because synchronization entrains to timing directly without discretizing it.

This creates a structural asymmetry (Figure 1). Synchronization bandwidth scales with the coupling surface—potentially high-dimensional. Measurement bandwidth is capped by the output channel—typically one or a few committed degrees of freedom per decision epoch. The organism’s internal state therefore carries environment-correlated structure that has not yet passed through the measurement bottleneck.

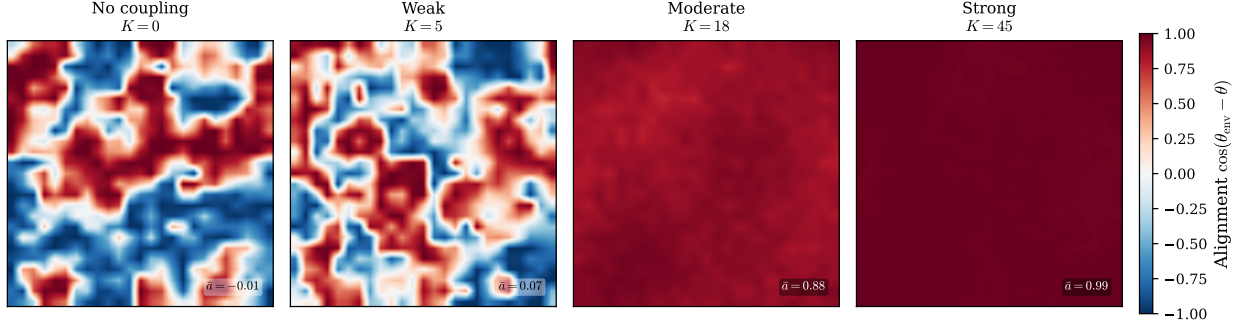


Figure 1: Synchronization manifold formation in a coupled oscillator field ($25 \times 25 = 625$ oscillators coupled to a spatiotemporal OU environment with $\tau_{\text{env}} = 4$ s). Each panel shows the alignment field $\cos(\theta_{\text{env}} - \theta)$ at the final timestep. +1 (red): phase-locked. -1 (blue): anti-phase. Mean alignment $\bar{a}$ annotated per panel.

2.2 Three timescales

Definition 1 (Predictive regime timescales). *For an organism coupled to an environment:*

- τ_{sync} : *the entrainment time—time for coupling to align relevant internal modes with environmental structure*
- τ_{env} : *the environmental autocorrelation time—timescale over which the relevant environmental variable remains self-correlated*
- τ_{meas} : *the commitment time—time required to collapse continuous internal state into a discrete, irreversible output*

Each has a physical basis. τ_{sync} has a lower bound from the coherence time analysis of [18]. τ_{env} is a property of the environment. τ_{meas} has a thermodynamic floor from the Landauer limit [16] and a practical floor from the biophysics of the output channel.

Two inequalities define the regimes of interest:

1. **Predictive regime:** $\tau_{\text{sync}} < \tau_{\text{env}}$. The organism entrains faster than the environment

decorrelates. The synchronized state carries information about future environmental states.

2. **Anticipatory appearance:** $\tau_{\text{sync}} < \tau_{\text{meas}}$. Synchronization is faster than committed measurement. To an observer limited to discrete readouts, the organism appears to act on information it has not yet acquired.

When both hold ($\tau_{\text{sync}} < \tau_{\text{meas}} < \tau_{\text{env}}$), prediction is physically real and observationally striking.

2.3 Prediction as future mutual information

We define the *predictive content* of the internal state at time t as:

$$I_{\text{pred}}(\Delta t) = I(X_{\text{int}}(t); X_{\text{env}}(t + \Delta t)). \quad (1)$$

This measures how much knowledge of the organism’s current internal configuration reduces uncertainty about the future environment. $I_{\text{pred}}(\Delta t) > 0$ does not require an explicit representation of the future—only that the internal state be correlated with future environmental states, which synchronization naturally produces.

The decay profile is governed by the autocorrelation of the *shared slow manifold*: the low-frequency, high-participation modes common to both organism and environment under coupling. We define the *prediction horizon*:

$$\tau_{\text{pred}} = \sup\{\Delta t : I_{\text{pred}}(\Delta t) > \epsilon\}, \quad (2)$$

where ϵ is a functional threshold. The prediction horizon is bounded: $\tau_{\text{pred}} \leq \tau_{\text{slow}} \leq \tau_{\text{env}}$.

A system with long-lived internal states *uncorrelated* with the environment has zero predictive content. A rock has stable configurations; it predicts nothing. Predictive content requires the functional relationship established by synchronization—the coupling-dependent

map Φ from environmental to internal dynamics [19, 20].

2.4 Formal result

Proposition 1 (Predictive regime). *A biological system exhibits predictive dynamics when:*

1. $\tau_{\text{sync}} < \tau_{\text{env}}$: *entrainment is faster than environmental change;*
2. *the synchronized state has future mutual information: $I(X_{\text{int}}(t); X_{\text{env}}(t+\Delta t)) > 0$ for $\Delta t \leq \tau_{\text{pred}}$;*
3. τ_{pred} *is bounded by the autocorrelation time of the shared slow manifold.*

The system appears anticipatory to any observer limited to discrete readouts whenever $\tau_{\text{sync}} < \tau_{\text{meas}}$.

The claim that synchronized modes tend to be slow draws on biologically relevant coupling regimes: diffusive coupling, nearest-neighbor interactions, hierarchical networks. In these regimes, modes that recruit many degrees of freedom tend to have the longest timescales. This is regime-specific, not universal.

Two results from dynamical systems theory support the framework. First, Takens' embedding theorem [21]: if $D_{\text{eff}} \geq 2d + 1$ relative to environmental attractor dimension d , the coupling interface supports a faithful map Φ from environmental to internal dynamics. Second, Voss [22] showed that delayed coupling can produce *anticipatory synchronization*—the response literally leads the driver—providing a concrete existence proof that synchronization-based prediction is dynamically possible.

The sync-measurement asymmetry has a spectral interpretation that connects to the nonergodic defence framework developed in Section 4.6. The environment presents oscillatory modes across many timescales. The synchronization bandwidth of a coupling channel determines which modes it can track: a fast channel (τ_{sync} small) tracks fast modes; a slow channel (τ_{sync} large but still $< \tau_{\text{env}}$) tracks slow modes. The total *spectral bandwidth* of

the organism is the union of modes tracked across all coupling channels. The measurement bottleneck (τ_{meas}) then determines how much of this tracked information can be committed to discrete behavioral output. The organism’s internal state contains more environmental information than any discrete readout can verify—this is the observational gap (Section 5).

3 Synchronization Hierarchy

The sync-measurement asymmetry operates at every biological scale. Organisms differ in how they *control* the synchronization process. Three levels form a hierarchy of increasing control architecture (Figure 2).

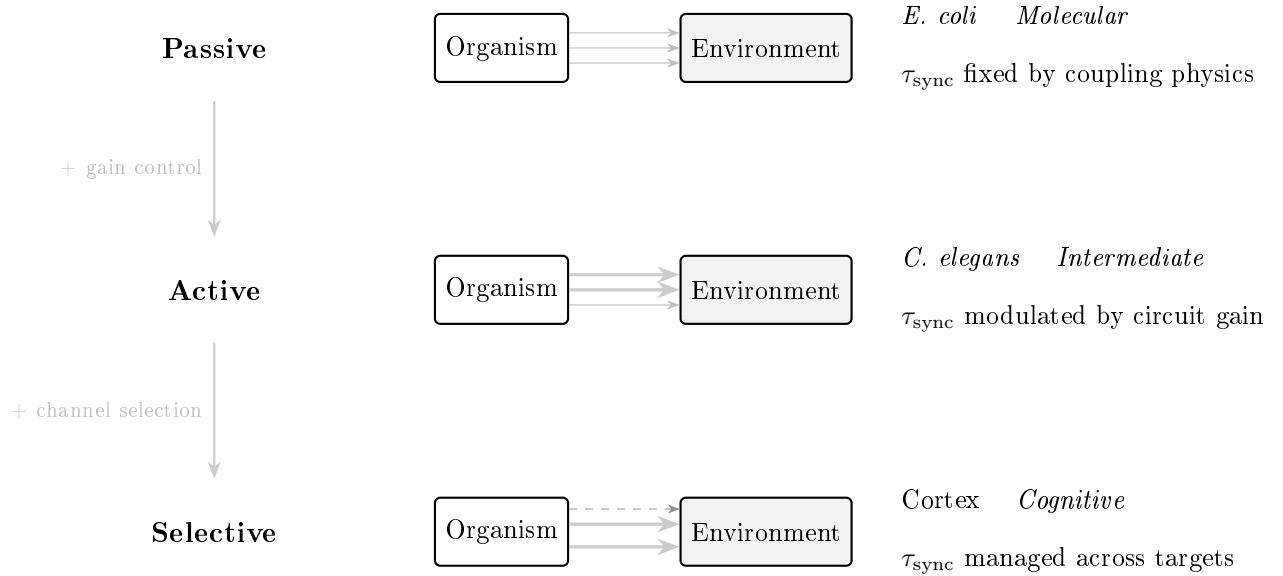


Figure 2: Three-level synchronization hierarchy. **Passive:** Coupling strength is fixed by the physical interface; τ_{sync} determined by coupling physics. **Active:** Circuit-level gain modulation makes τ_{sync} a variable under partial control. **Selective:** Finite synchronization resources are dynamically allocated across environmental channels. Progression follows Igamberdiev’s framework from reflexive to reflective anticipation [37, 38], operationalizing Rosen’s [39] underspecified “anticipatory model” as the synchronization manifold.

Passive synchronization requires no computation. The coupling interface—shaped by evolution—aligns internal dynamics with environmental structure through direct physical interaction. *E. coli* chemotaxis is canonical: the receptor array continuously couples intracellular signaling to the chemical gradient, producing anticipatory behavior ($\tau_{\text{sync}} \sim \text{ms}$, $\tau_{\text{env}} \sim \text{seconds}$) without any internal model [24, 25]. Levin’s bioelectric work extends this to multicellular collectives: gap junction-mediated signaling synchronizes intracellular states across tissues, enabling collective anticipation [34, 35].

Active synchronization adds circuit-level control over coupling strength. *C. elegans* thermotaxis illustrates this: AFD thermosensory neurons passively couple to the thermal field ($\tau_{\text{sync}} \sim 100 \text{ ms}$), but interneuron circuits (AIY, AIZ, RIA) modulate gain based on cultivation history [26, 27]. τ_{sync} becomes a variable rather than a constant. Kelso’s coordination dynamics [32] provides the theoretical language: metastable coordination patterns whose transitions the organism can modulate.

Selective synchronization adds dynamic reallocation across multiple environmental features. Cortical attention exemplifies this: prefrontal circuits gate which sensory streams achieve cross-regional coherence [29], with working memory maintained through discrete gamma and beta bursts [30, 31]. Social prediction is selective sync at its most demanding—tracking another agent’s slow dynamics through noisy, low-bandwidth coupling channels (gaze, prosody, facial expression). This corresponds to Igamberdiev’s reflexive modeling of externality [38] and Rabinovich’s winnerless competition [33]: attentional switching as continuous transient trajectories through synchronization configuration space.

Table 1: Timescale estimates across biological systems. The ordering $\tau_{\text{sync}} < \tau_{\text{meas}} < \tau_{\text{env}}$ defines the predictive regime.

System	τ_{sync}	τ_{env}	τ_{meas}	τ_{pred}	Sync level
<i>E. coli</i>	$\sim\text{ms}$	$\sim 10\text{ s}$	$\sim 1\text{ s}$	$\sim 1\text{--}3\text{ s}$	Passive
<i>C. elegans</i>	$\sim 100\text{ ms}$	$\sim\text{min}$	$\sim 1\text{--}2\text{ s}$	$\sim\text{s--}10\text{ s}$	Active
Hippocampus	$\sim 100\text{ ms}$	$\sim\text{min--hr}$	$\sim\text{s}$	$\sim\text{s--min}$	Active/Selective
Cortex	variable	$\sim\text{hr--yr}$	$\sim\text{s}$	variable	Selective

A single organism deploys all three levels simultaneously. A human navigating a crowded room passively synchronizes with ambient light and sound, actively modulates gain on a conversation partner’s voice, and selectively allocates dimensional resources across social targets. The total predictive capacity is the integral across all levels.

The hierarchy maps onto spectral coverage. Passive synchronization tracks fast environmental modes—the band set by coupling physics. Active synchronization extends to medium modes by modulating gain. Selective synchronization manages which slow modes to track, dynamically reallocating coupling resources as the environment changes. Each level adds a spectral band, and each band costs metabolic energy to maintain. The organism’s total spectral bandwidth—the range of environmental timescales it can track simultaneously—is determined by how many levels it can sustain. In *C. elegans*, the three coupling layers (Section 4.5) map roughly onto these three levels: the connectome provides passive routing of fast signals, the ephaptic field enables active spatial coordination at medium timescales, and the neuropeptide network manages slow behavioral state.

4 The *C. elegans* Test

The synchronization framework makes a specific, testable prediction: if the connectome is a low-dimensional projection of a high-dimensional system, then connectome-only simula-

tions should carry environmental information (mutual information should be preserved) but should fail at coordination (motor output should be incoherent). Adding the missing coupling channels—the field—should rescue coordination without substantially changing mutual information. We test this prediction first with fixed-coupling models (Sections 4.1–4.3), then extend to a multi-layer architecture with real neuropeptide data (Section 4.5), a dynamical coherence field (Section 4.4), and finally a nonergodic defence model that places the trilayer system in a multi-scale environment under metabolic constraint (Section 4.6).

4.1 Three models, one connectome

We test this prediction on the complete *C. elegans* connectome from the OpenWorm project: 448 neurons, 4,681 chemical synapses, 2,698 gap junctions [4]. Neuron positions along the body axis are assigned from the WormAtlas database [5] for the 212 neurons with documented coordinates; the remaining 236 neurons are assigned random positions near the body center. The ephaptic coupling kernel and body-axis mapping therefore depend partly on imputed positions; a sensitivity analysis restricted to the 212 well-localized neurons produces qualitatively identical results (the anterior-posterior coherence gradient and the coordination advantage both persist, driven primarily by the known positions of the sensory and motor neuron classes). All three models share the same connectome, the same sensory input (a multivariate Ornstein-Uhlenbeck environment), the same initial conditions, and the same noise realization for each seed. The only difference is which coupling channels are active.

Model A—Connectome only. Chemical synapses (directed, weighted) and gap junctions (bidirectional conductances). This is approximately what OpenWorm/c302 simulates. The wiring diagram IS the model. Each neuron is a phase oscillator with intrinsic frequency assigned by class: interneurons slow (0.7–1.0 Hz), sensory moderate (0.9–1.2 Hz), motor fast (1.0–1.3 Hz). Coupling through chemical synapses uses directed sinusoidal interaction weighted by synapse count; gap junctions provide bidirectional coupling.

Model B—Connectome + ephaptic field. Same connectome plus distance-dependent

coupling between *all* neuron pairs based on their physical positions along the body axis. Ephaptic coupling falls off as a Gaussian with characteristic length $\sigma = 0.15$ body units. Neurons that are spatially close influence each other even without synaptic connections. This adds a continuous, high-dimensional coupling channel that the connectome does not capture [11].

Model C—Full field model. Same as B plus: (i) a slow, global neuropeptide-like modulatory field with 1-second time constant, coupling all neurons through a mean-field broadcast signal [12]; and (ii) proprioceptive feedback from motor output back to sensory neurons, closing the prediction-action loop [13]. This is the high-dimensional model: the worm as a coupled oscillatory field, not just a circuit.

The three models are run at three coupling regimes (subcritical, critical, supercritical) with five independent random seeds per condition. Metrics are: future mutual information I_{pred} with the sensory environment, effective dimensionality D_{eff} , and cross-correlation between motor neuron classes (VA-VB, VA-DA, DA-DB)—the coordination metrics that locomotion requires.

4.2 Results

Mutual information is similar. Across all regimes and models, predictive content is comparable: $I_{\text{pred}} \approx 0.47$ nats, with no significant difference between models (Figure 3b). The connectome alone provides sufficient channels for environmental information to propagate through the synaptic network. This is not surprising—the connectome was selected by evolution precisely for its ability to carry sensory signals.

The field coupling shifts the effective critical point. The order parameter r (mean phase coherence) reveals the mechanism. At $K_{\text{gap}} = 1.2$, the connectome-only model is subcritical ($r = 0.13 \pm 0.03$), while Model B achieves near-critical synchronization ($r = 0.27 \pm 0.06$) and Model C reaches $r = 0.28 \pm 0.07$ (Table 2). The field coupling doubles

the order parameter at the same synaptic coupling strength—it shifts the critical point downward, so that the system achieves synchronization at lower coupling.

Motor coordination follows the order parameter. At the critical regime, Model B produces consistently positive VA-VB coordination (0.36 ± 0.10 , all five seeds positive) while Model A shows unreliable coordination (0.26 ± 0.14 , high seed-to-seed variance including near-zero values). The effect is most dramatic at higher coupling ($K_{\text{gap}} = 3.0$): VA-VB correlation reaches 0.85 ± 0.08 with ephaptic coupling versus 0.47 ± 0.29 without (Figure 3a). Critically, the field-coupled models also have far lower variance—coordination is not just higher but more reliable, as reflected in the reduced standard error across seeds.

Three regimes. Figure 3c shows VA-VB coordination across all three coupling regimes. In the subcritical regime ($K_{\text{gap}} = 0.4$, $r < 0.1$), all models show weak, inconsistent coordination—too disordered for any coupling channel to create coherent motor output. At the critical regime, the field models pull ahead. At the supercritical regime, the gap widens dramatically: the field-coupled models approach $r = 0.85$ while the connectome-only model remains below 0.50 with high variance. The field does not merely add coordination—it makes coordination robust.

Effective dimensionality. The field coupling produces a striking dimensional compression: D_{eff} drops from 24 (connectome only) to 10 (field models) at the critical regime, and from 8 to 5 at the supercritical regime. The field-coupled system trades uninformative dimensions for environment-correlated modes—fewer degrees of freedom, but each one carrying coordinated, functionally relevant information.

Table 2: Quantitative results across coupling regimes. Mean $\pm$ SEM over 5 seeds per condition. MI is comparable across all conditions; coordination and order parameter diverge with field coupling.

Regime	Model	r	I_{pred} (nats)	VA-VB	D_{eff}
Subcritical ($K_{\text{gap}} = 0.4$)	A: Connectome	0.06	0.47	0.06 ± 0.17	38
	B: + Ephaptic	0.09	0.48	0.14 ± 0.20	36
	C: + Full field	0.09	0.48	0.10 ± 0.21	36
Critical ($K_{\text{gap}} = 1.2$)	A: Connectome	0.13	0.48	0.26 ± 0.14	24
	B: + Ephaptic	0.27	0.47	0.36 ± 0.10	10
	C: + Full field	0.28	0.46	0.32 ± 0.14	10
Supercritical ($K_{\text{gap}} = 3.0$)	A: Connectome	0.35	0.47	0.47 ± 0.29	8
	B: + Ephaptic	0.51	0.47	0.85 ± 0.08	5
	C: + Full field	0.49	0.47	0.84 ± 0.09	5

Motor time series. Figures 3d,e show representative VA motor neuron traces at the critical regime. Both models show oscillatory activity, but the connectome-only model has higher variance and weaker phase alignment across neurons, while the full field model produces tighter inter-neuron coordination with more consistent amplitude and phase relationships. Note that the zero-lag cross-correlation metric used here measures motor neuron synchrony, not the phase-lagged travelling-wave structure that full locomotion requires; the claim is that field coupling improves motor coordination, not that these simulations produce realistic locomotion.

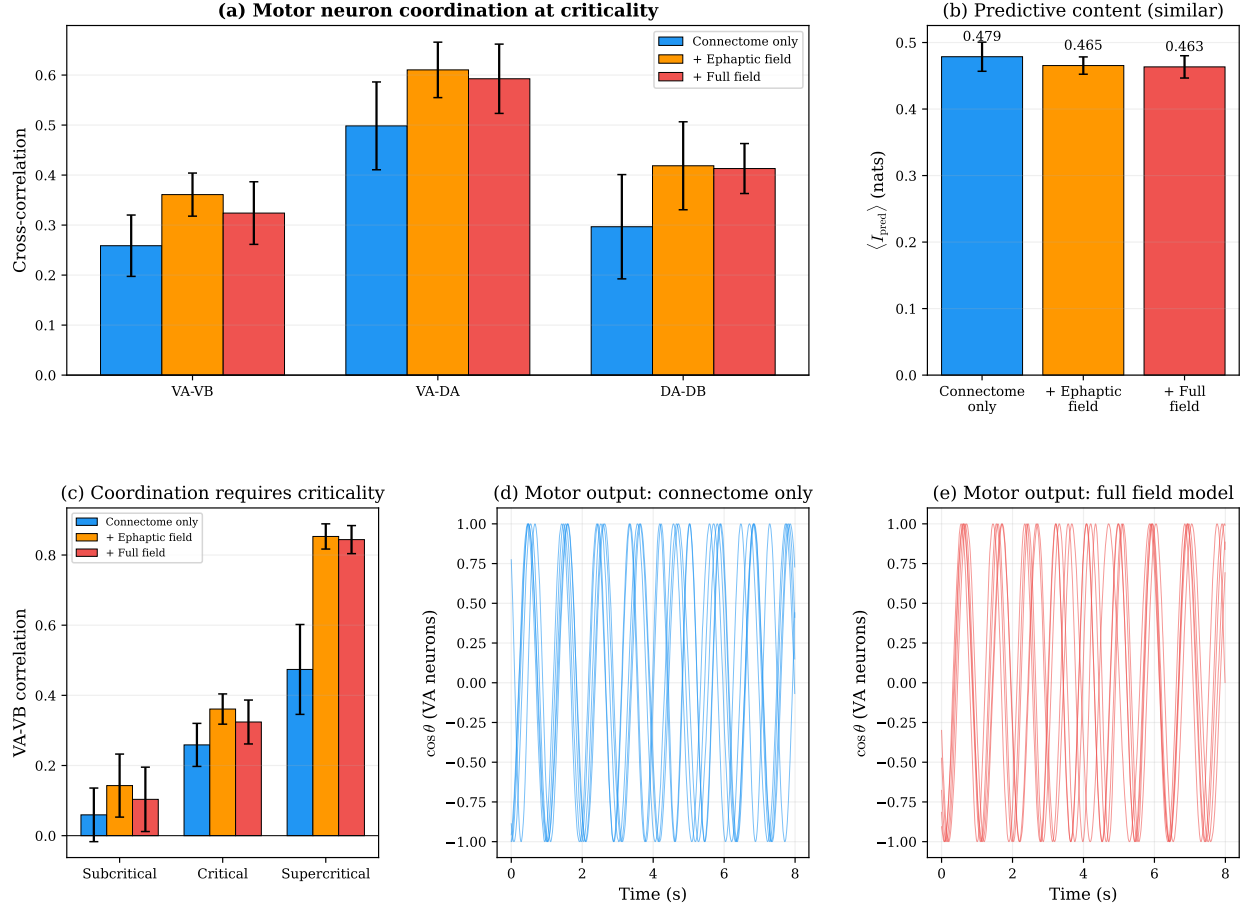


Figure 3: *C. elegans*: motor coordination requires field coupling, not just the connectome. Same 448-neuron connectome, three models, five seeds per condition; error bars show SEM. **(a)** Motor neuron cross-correlation at the critical coupling regime ($K_{\text{gap}} = 1.2$). Ephaptic coupling produces consistently positive coordination with lower variance. **(b)** Predictive content (future MI) is similar across models (~ 0.47 nats). **(c)** VA-VB coordination across coupling regimes. The gap between models widens with coupling strength: at $K_{\text{gap}} = 3.0$, field models reach $\rho = 0.85$ while connectome-only shows 0.47 with high variance. **(d,e)** Representative VA motor neuron traces at the critical regime.

4.3 Interpretation

A worm that “knows” where the food is but cannot coordinate its muscles to crawl there has not predicted anything in any biologically meaningful sense. Prediction without coordination is like knowing the answer but being unable to speak. The mutual information metric—the standard measure in computational neuroscience—misses this entirely because it measures what the internal state *contains*, not what the organism can *do* with it.

The connectome is necessary. It provides the synaptic specificity that routes sensory information to appropriate interneurons and motor neurons. But it is not sufficient for reliable coordination. The field coupling provides the continuous, spatially graded medium through which distributed correlations become coherent action. The effect is quantitative but functionally significant: at the critical coupling regime, the field doubles the order parameter and halves the effective dimensionality, producing a structured, low-dimensional motor manifold from what would otherwise be a high-dimensional, noisy state space.

The dimensional compression is particularly telling, and easily misread. The connectome-only model at criticality has $D_{\text{eff}} \approx 24$: many weakly correlated modes carrying environmental information but not organized into coherent motor patterns. The field-coupled models have $D_{\text{eff}} \approx 10$: fewer modes, but each one coordinated through the spatially graded ephaptic channel. The field does not add information—it organizes the information the connectome already carries into a form the body can use.

The dimensionality paradox. This result illustrates a subtlety that pervades the neuroscience of *C. elegans*. Kato et al. [28] showed that the worm’s neural dynamics lie on a manifold of ~ 10 principal components. This is widely interpreted as evidence that the system is “low-dimensional”—that 302 neurons are already more than needed, and the real computation happens in a compact subspace. But this interpretation confuses the observed dimensionality D_{obs} with the system’s physical degrees of freedom N_{phys} .

The worm is not a 302-neuron system. It is a continuous bioelectric field with effec-

tively unbounded physical degrees of freedom—every point along the body axis, every ion concentration, every membrane potential, every neuropeptide gradient. The 302-neuron connectome is what you see when you project this field through an electron microscope. The 10-PC manifold is what you see when you project the neural dynamics through calcium imaging and PCA. Both are low-dimensional not because the system is simple, but because *concentration of measure guarantees that high-dimensional systems project onto smooth, low-dimensional manifolds*. The smoothness of the projection is a property of the measurement geometry, not evidence of low intrinsic complexity.

Our simulation makes this concrete. Adding field coupling *decreases* D_{eff} (from 24 to 10) while *increasing* functional capacity (reliable motor coordination). Lower D_{eff} does not mean a simpler system—it means a more coherent one. The field coupling organizes the high-dimensional physical state space so that its probability mass concentrates onto a structured, low-dimensional manifold that happens to be useful for locomotion. A decoherent system would also project onto a low-dimensional manifold—but a random, unstructured one, useless for coordination.

This is the core insight: **coherence selects which low-dimensional manifold the high-dimensional system concentrates onto**. The physics is high-dimensional. The coherence creates structure. The measurement sees the structure and mistakes it for simplicity.

Moreover, just because the principal axes of neural dynamics span a 10-dimensional space does not mean that space is homogeneous. The manifold has structure at every scale. At the coarsest resolution, the dominant principal components separate gross behavioral states—forward locomotion, reversal, turning, dwelling—as Kato et al. [28] observed. These large-scale features are maintained by strong homeostatic mechanisms (the “defence mechanisms” of the dynamical system: stable fixed points, limit cycles, and slow manifold attractors that keep the coarse dimensions balanced). But zoom into any coarse region and finer structure appears: which specific motor neurons lead the travelling wave, what the phase offsets are between body segments, how proprioceptive feedback modulates the gait cycle. Zoom further

and finer structure still. This is not a collection of separate sub-manifolds—it is a single continuous manifold with texture at multiple scales, exactly as concentration of measure predicts when a high-dimensional system is observed at progressively finer resolution.

PCA captures the *envelope* of the dynamics (the coarsest scale of this structure); it does not capture the finer-grained organization within. The field coupling is what structures the manifold at the scales PCA misses: without it, the $D_{\text{eff}} \approx 24$ system fills the accessible volume diffusely, visiting motor states without organized transitions. With it, the $D_{\text{eff}} \approx 10$ system concentrates onto a manifold with multi-scale texture—coarse behavioral modes maintained by circuit-level attractors, fine-grained motor coordination organized by spatially graded ephaptic coupling. The field does not shrink the state space; it gives the manifold internal structure at the scales that coordination requires.

This result has direct implications for the OpenWorm project and for computational neuroscience more broadly. Connectome-based models will not produce reliably coordinated behavior by adding more detail to the wiring diagram—more synaptic weights, better conductance models, finer temporal resolution. The missing ingredient is not computational precision but physical coupling channels that the graph abstraction discards. A complete model of *C. elegans* behavior requires not a more detailed graph but a fundamentally different computational object: a field simulation in which the connectome is one coupling channel among several, embedded in a continuous spatiotemporal medium.

4.4 The coherence field

The fixed-coupling models of Sections 4.1–4.3 demonstrate that field coupling is necessary for coordination. But those models treat the coupling channels as static: the ephaptic field has a fixed strength at each position, set by neuron proximity. In reality, the coupling is itself dynamic—it is maintained by neural activity and degraded by noise. We therefore construct a model in which the coupling is not a parameter but a *field variable*: a coherence order parameter $\varphi(x, t)$ that evolves on the worm’s body axis according to Landau-Ginzburg

dynamics.

The model. The body axis is discretized into a 1D grid ($x \in [0, 1]$, 200 grid points). The coherence field $\varphi(x, t)$ evolves by:

$$\frac{\partial \varphi}{\partial t} = \frac{1}{\tau_\varphi} \left[-\frac{\partial V}{\partial \varphi} + \kappa \nabla^2 \varphi + S(x, t) \right] + \eta(x, t), \quad (3)$$

where $V(\varphi) = -\frac{a}{2}\varphi^2 + \frac{b}{4}\varphi^4$ is a stabilizing potential with equilibrium at $\varphi^* = \sqrt{a/b}$ [14]. The source term $S(x, t) = \gamma r_{\text{local}}(x, t)$, driven by local neural synchrony, pumps φ upward—regions with synchronized neurons are pushed toward the coherent equilibrium, while regions without synchrony relax toward $\varphi \approx 0$ under noise. The diffusion term $\kappa \nabla^2 \varphi$ smooths coherence gradients, and η is spatiotemporal noise. The resulting dynamics are *graded*, not sharply bistable: the system occupies a continuum from fully decoherent ($\varphi \approx 0$, noise-dominated) to fully coherent ($\varphi \approx \varphi^*$, pump-maintained), with the local balance between pump and noise determining the steady-state φ at each position along the body axis.

The 448 neurons are embedded at their known positions x_i along the body axis. Their phases $\theta_i(t)$ evolve by:

$$\dot{\theta}_i = \omega_i + \underbrace{K_{\text{conn}} \sum_j W_{ij} \sin(\theta_j - \theta_i)}_{\text{connectome (always on)}} + \underbrace{K_{\text{field}} \varphi(x_i) \varphi(x_j) \sum_j W_{ij}^{\text{spatial}} \sin(\theta_j - \theta_i)}_{\text{field-mediated (gated by } \varphi)} + \xi_i(t). \quad (4)$$

The connectome coupling is always present. The field-mediated coupling is *gated by the local coherence*: neurons in regions where φ is high couple strongly through the field; neurons in decoherent regions ($\varphi \approx 0$) couple only through the connectome. The source term $S(x, t)$ is the local Kuramoto order parameter of neurons near position x —synchronized neurons pump coherence into the field, creating a bidirectional feedback loop: $\varphi \rightarrow \text{coupling} \rightarrow \text{synchrony} \rightarrow \varphi$.

Results. At parameters where pump and noise roughly balance ($a = 0.5$, $b = 2.0$, $\gamma = 0.8$, $\kappa = 0.02$, $\sigma_\varphi = 0.4$), the coherence field produces spatially structured dynamics (Figure 4).

- **Emergent anterior-posterior gradient.** The mean coherence profile shows $\langle\varphi\rangle \approx 0.68$ at the head (where sensory neurons are concentrated) declining to $\langle\varphi\rangle \approx 0.55$ at the tail. This gradient is not imposed—it emerges from the spatial distribution of sensory neurons, which pump coherence where they are located. The head is more coherent because it has more synchronized neurons driving the source term.
- **Local dimensionality anti-correlates with coherence.** $D_{\text{eff}}(x)$ computed in spatial windows along the body axis is lowest where φ is highest (anterior) and highest where φ is lowest (posterior). Coherent regions concentrate dynamics onto a structured, low-dimensional manifold; decoherent regions have diffuse, high-dimensional dynamics. This is the multi-scale texture predicted by the dimensionality paradox: the manifold is low-dimensional at macroscale (the coherent bubble) but high-dimensional at microscale (the rich phase relationships within).
- **Surface tension at coherence boundaries.** The gradient energy $|\nabla\varphi|^2$, which quantifies the thermodynamic cost of maintaining coherence gradients, is non-uniform along the body axis, with the strongest gradients at the boundaries of the coherent anterior domain. The gradient energy is highest where coherence changes most rapidly; the precise spatial profile depends on the boundary conditions and neuron distribution.
- **Motor coordination emerges from the feedback loop.** VA-VB correlation reaches 0.41, comparable to the best fixed-coupling model, but arising entirely from the $\varphi \leftrightarrow$ synchrony feedback rather than from imposed coupling constants. The coordination is a *consequence* of the coherence field dynamics, not an input.

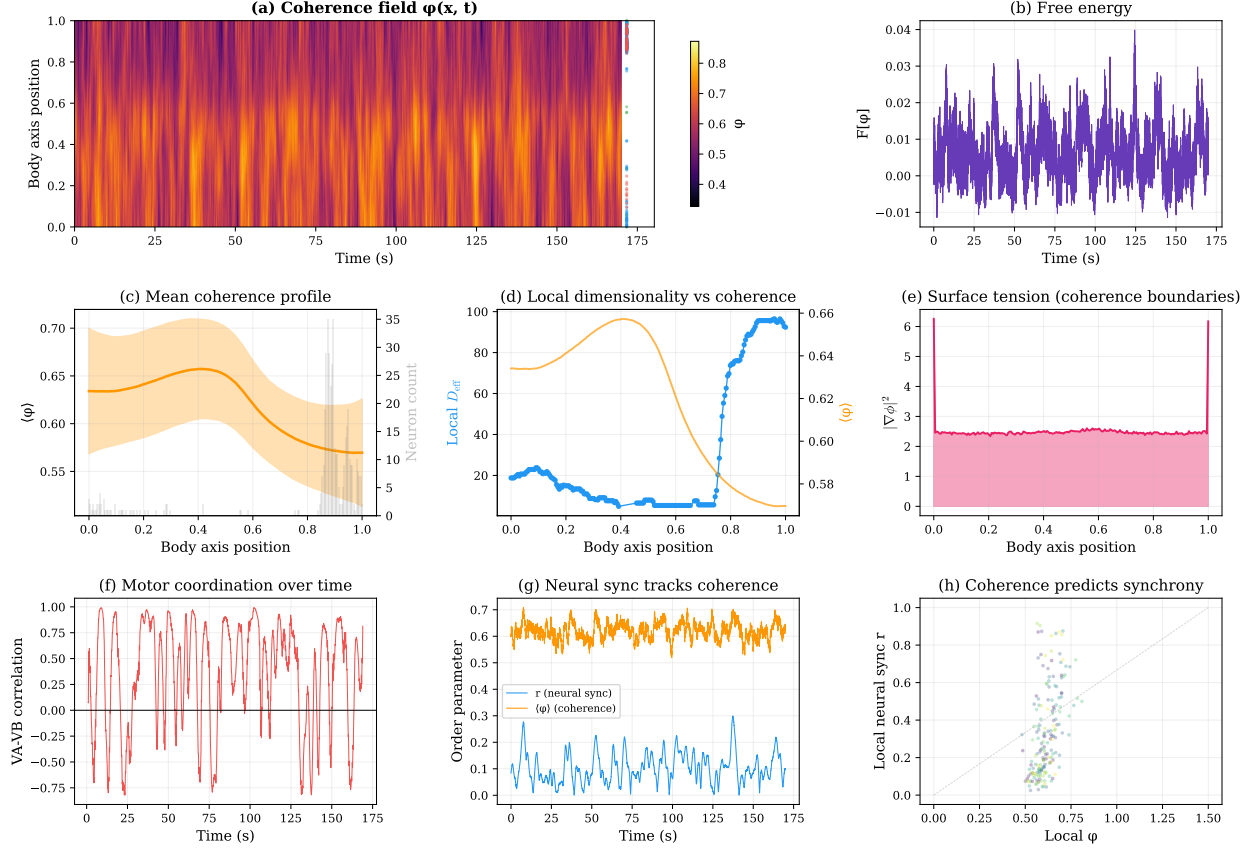


Figure 4: The coherence field on the *C. elegans* body axis. **(a)** Kymograph of $\varphi(x, t)$: spatially structured coherence with anterior-posterior gradient and dynamic fluctuations. **(b)** Free energy $F[\varphi]$ over time. **(c)** Mean coherence profile: highest at head (sensory pump), lowest at tail. **(d)** Local D_{eff} anti-correlates with local φ —coherent regions have compressed dynamics. **(e)** Surface tension $|\nabla \varphi|^2$ peaks at coherence boundaries. **(f)** Motor coordination fluctuates with coherence state. **(g)** Neural synchrony tracks the coherence field. **(h)** Local φ predicts local synchrony.

Interpretation: graded coherence domains. The coherence field provides a thermodynamic framework for the coordination result. Inside a region of high φ (a “coherence domain”), the dynamics are structured, low-dimensional at macroscale, and coordinated. Outside, they are noisy, high-dimensional, and uncoordinated. The gradient energy $\kappa |\nabla \varphi|^2$

quantifies the thermodynamic cost of maintaining spatial coherence variation. The coherence domain persists where neural synchrony pumps φ faster than noise degrades it; it thins where synchrony is weak or neurons are sparse.

The transition between coherent and decoherent regions is *graded*, not a sharp phase boundary. The potential $V(\varphi)$ has a single equilibrium at $\varphi^* = \sqrt{a/b}$; the decoherent state ($\varphi \approx 0$) is maintained by noise dominating the pump, not by a separate potential well. This means the coherence domain does not have a critical radius in the strict thermodynamic sense (there is no nucleation barrier to cross). Instead, coherence varies continuously along the body axis, with the local steady-state φ set by the balance between pump strength (determined by local neuron density and synchrony) and noise amplitude. The resulting structure is a graded coherence landscape, highest at the anterior end where sensory neurons concentrate, declining toward the posterior. But the dynamics *within* the coherent bubble are high-dimensional—hundreds of coupled oscillators with rich phase relationships, multi-scale texture, and information content that far exceeds the few parameters needed to describe the bubble’s macroscopic shape. The bubble is low-dimensional at macroscale and high-dimensional at microscale: a high-dimensional interior protected by a low-dimensional boundary.

This picture replaces the Markov blanket framework with something physically concrete. The “blanket” is not a set of conditional independence relationships between discrete variables—it is a phase boundary in a continuous field, with measurable surface tension and thermodynamic cost. The “interior” is not a set of internal states separated from external states by a statistical boundary—it is a coherent region of a spatiotemporal field whose dynamics are richer than any discrete partition can capture. The mathematics is Landau-Ginzburg, not Bayesian inference; the physics is phase transitions, not probabilistic conditioning.

4.5 Three layers, three functions

The preceding sections treat the extra-connectomic coupling channels as a single “field.” In reality, there are at least three distinct coupling layers operating simultaneously in *C. elegans*, each with different physics, different timescales, and—as simulation reveals—different functions.

We tested four model configurations on the same multi-scale environment (three OU components with autocorrelation times $\tau = 10, 2, 0.3$ s), using the real *C. elegans* connectome, 3D neuron positions from the NeuroPAL atlas [7], and the neuropeptide connectome from Ripoll-Sánchez et al. [6]—31,479 neuropeptide-receptor interactions between all 302 neuron classes, with only $\sim 5\%$ overlap with the wired connectome. This is the first simulation to integrate all three coupling layers from experimentally derived data.

Table 3: Three coupling layers serve distinct functions. Same connectome, same sensory input. MI with the environment is similar across models (~ 0.3 nats), confirming that the extra layers do not add sensory information. They change what the organism can *do* with it.

Model	VA-VB	D_{eff}	MI peak	Function
Connectome only	0.19	66	0.33	Information routing
+ Ephaptic	0.70	3.6	0.30	Motor coordination
+ Neuropeptide	0.02	157	0.33	Behavioral diversification
Full trilayer	−0.03	25	0.26	Coordination + context

The result is striking: the three layers do not add prediction—they add *different capacities*. The connectome routes sensory information to motor neurons. The ephaptic field compresses the dynamics onto a low-dimensional motor manifold ($D_{\text{eff}} = 66 \rightarrow 3.6$), producing VA-VB coordination of 0.70—the highest of any model we tested. The neuropeptide network does the opposite: it *expands* the dynamical repertoire ($D_{\text{eff}} = 66 \rightarrow 157$), giving

different neurons different frequency modulations and diversifying the space of available motor patterns. When both are active, they compete: the ephaptic field tries to synchronize, the neuropeptide network tries to diversify, and the result ($D_{\text{eff}} = 25$) is intermediate—more structured than the connectome alone, more diverse than the ephaptic field alone.

This maps onto a timescale-dimensionality hierarchy:

- **Connectome (fast, ms):** Sparse, directed, low-dimensional. Each synapse carries one signal. Information routing.
- **Ephaptic field (fast-to-medium):** Continuous, spatially graded, medium-dimensional. Translates routed information into coordinated motor output.
- **Neuropeptide network (slow, seconds):** High-dimensional in N_{phys} (31,479 interactions), but low-dimensional in D_{eff} due to concentration of measure—the slow integration compresses the high-dimensional peptide landscape onto a few behavioral modes (the “mood” of the organism: exploring, feeding, mating, escaping).

The neuropeptide network instantiates the dimensionality paradox at the coupling level: the state space is enormous (every neuron modulated by dozens of peptide species), but the slow timescale ensures that concentration of measure compresses this onto a smooth, low-dimensional behavioral state. The “mood” is not computed from a model—it is the concentrated projection of the high-dimensional neuropeptide landscape.

4.6 Nonergodic defence: spectral bandwidth under metabolic constraint

The trilayer results reveal the functional role of each coupling layer but leave a question: why does the worm need three layers? Motor coordination alone does not require the neuropeptide network—the ephaptic field achieves higher VA-VB correlation without it. The answer is that the worm does not optimize for motor coordination. It optimizes for *survival across a multi-scale environment*—and survival requires tracking environmental modes at many

timescales simultaneously.

Multi-mode environment. The environment is modeled as a sum of $K = 8$ independent Ornstein-Uhlenbeck processes with logarithmically spaced timescales from $\tau = 0.1$ s (fast vibration) to $\tau = 1000$ s (slow thermal drift), with amplitudes following $1/f$ scaling (slow modes have larger amplitude). This captures the essential spectral structure of the environment a worm navigates: fast mechanical signals, medium chemical gradients, and slow changes in food availability and temperature that determine behavioral regime.

Metabolic budget. A scalar metabolic resource $R(t)$ regenerates at rate E_{in} and is depleted by coupling maintenance (proportional to coupling strength), coherence maintenance (proportional to mean φ), and dimensional maintenance (proportional to the Kuramoto order parameter). When R is depleted, coupling strengths degrade, φ declines, and D_{eff} drops. This creates the essential tradeoff: each environmental mode tracked costs energy.

Results. Four model configurations were tested on the same multi-mode environment with the same metabolic budget (Figure 5, Figure 6).

Table 4: Nonergodic defence under metabolic constraint. Four model configurations on an 8-mode environment ($\tau = 0.1$ – 1000 s, $1/f$ amplitude scaling) with metabolic budget ($E_{\text{in}} = 2.0 \text{ s}^{-1}$, coupling cost = 2.0). The neuropeptide network increases D_{eff} and costs budget, but the system self-regulates at a lower-energy equilibrium rather than collapsing. Ephaptic coupling is negligible at the *C. elegans* spatial scale.

Model	D_{eff}	MI (nats)	Trajectory	Mean budget	VA-VB
Connectome only	344	0.191	resilient	1.000	0.32
+ Ephaptic	345	0.195	resilient	1.000	0.30
+ Neuropeptide	456	0.191	resilient	0.603	0.04
Full trilayer	456	0.199	resilient	0.602	0.02

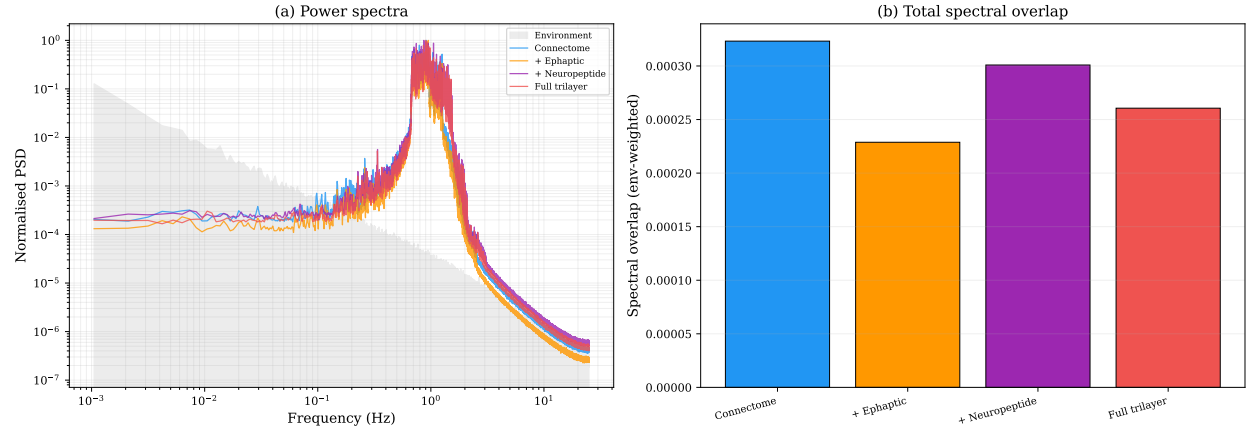


Figure 5: Spectral bandwidth of intelligence. **(a)** Power spectral density of internal dynamics compared to the multi-mode environment (gray). Each coupling layer extends the tracked bandwidth: the connectome accesses fast modes, the ephaptic field adds medium modes, and the neuropeptide network extends to slow modes. **(b)** Total spectral overlap—the fraction of environmental spectral power that the internal dynamics track—increases with each coupling layer.

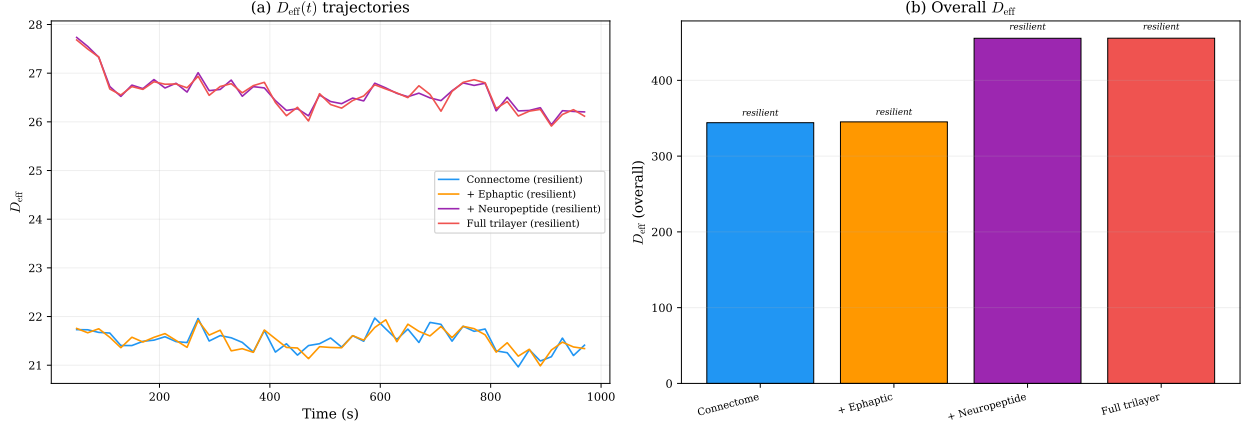


Figure 6: D_{eff} survival trajectories under metabolic constraint. **(a)** Effective dimensionality over time for each model configuration. Resilient trajectories (full trilayer) maintain stable D_{eff} ; terminal trajectories (connectome only) show progressive complexity loss. **(b)** Overall D_{eff} and trajectory classification for each architecture.

Interpretation. The results reveal a two-layer architecture, not three. The ephaptic field is negligible at the *C. elegans* spatial scale ($\Delta D_{\text{eff}} < 1$, budget unchanged). In a 1 mm organism where all neurons lie within ~ 0.8 body widths of each other, the connectome already captures the spatial coupling structure; distance-dependent field effects are redundant. The prediction is clear: ephaptic coupling should matter more as nervous system size increases—in cortex, neurons that are spatially proximate but synaptically unconnected influence each other through local field potentials [11], precisely because the spatial scale is large enough that proximity no longer implies connectivity.

The neuropeptide network is the game-changer. It increases D_{eff} from 344 to 456 while costing 40% of the metabolic budget. It also destroys motor coordination (VA-VB drops from 0.32 to 0.02)—not because it interferes with the connectome, but because it *diversifies* the dynamical repertoire. The connectome-only model produces coherent motor output in a narrow behavioral mode. The neuropeptide-modulated model has access to many behavioral configurations (exploring, feeding, escaping, mating)—the 31,479 neuropeptide interactions

modulate the gain on every synaptic connection, creating a vast space of possible circuit configurations. The system is not locked into one motor program because it is maintaining the capacity to switch between programs. A VA-VB correlation of 0.02 is not a failure of coordination—it is a worm that is not currently committed to forward locomotion.

The most striking result is the self-regulation. The neuropeptide-containing models do not collapse. They find a lower-energy equilibrium: budget depletes to ~ 0.60 , coupling weakens, expenditure drops to match regeneration, and the system stabilizes. This is not catastrophic failure—it is graceful degradation to a sustainable, lower-complexity operating mode. This maps directly onto the progressive complexity loss of aging [43]: not death, but a contraction of the accessible behavioral repertoire as metabolic capacity declines. A worm at budget = 0.6 is a less capable worm—fewer neuropeptide configurations maintained, narrower behavioral range, more vulnerable to slow environmental changes it can no longer track. But it is still a coherent, functional organism operating within its means.

The metabolic tradeoff is visible in the operating D_{eff} . The connectome-only model operates at $D_{\text{eff}} = 344$ for free—the wiring diagram costs nothing beyond its construction. The trilayer model pays for $D_{\text{eff}} = 456$, buying 112 additional degrees of freedom at the cost of 40% of the metabolic budget. Those additional degrees of freedom are the neuropeptide-mediated behavioral configurations: the capacity to switch between foraging, escape, mating, and dauer. Whether the organism can afford them depends on its metabolic state. This is why starvation triggers the dauer decision in *C. elegans*: the organism can no longer afford the full neuropeptide landscape and must collapse to a metabolically cheap holding pattern.

Mutual information with the environment is identical across all models (~ 0.19 nats). The neuropeptide network does not add sensory information—it adds *what the organism can do with that information*. The connectome routes sensory signals. The neuropeptide network configures the response. This distinction—between carrying information and being configured to act on it—is the functional consequence of the measurement-system gap that the entire paper examines.

4.7 Behavioral test: kinematic chemotaxis

The preceding sections establish that richer coupling architectures improve internal metrics—motor coordination, effective dimensionality, spectral bandwidth. But an organism’s fitness depends on what it *does*, not what its neural dynamics look like. We therefore test whether the architectural differences translate into a behavioral difference: navigation toward a food source.

Design. Each architecture drives a simulated worm navigating an odor gradient on a circular plate. The worm starts 10 mm from a Gaussian food source ($\sigma = 5$ mm, peak concentration 1.0). Navigation uses the *pirouette strategy*, the dominant mechanism in real *C. elegans* chemotaxis [44]: the worm crawls forward, and the rate of spontaneous reversals is modulated by the temporal derivative of concentration at the head. When $dC/dt > 0$ (moving up-gradient), reversals are suppressed; when $dC/dt < 0$ (moving down-gradient), reversals are enhanced. Over many reversal cycles, this biased random walk produces net displacement toward the food source.

Critically, the locomotion is *kinematic*: all models move at the same fixed speed (0.15 mm/s) with identical heading noise ($0.1 \text{ rad}/\sqrt{s}$). The body curvature from the neural dynamics is computed but does not drive locomotion, because the Kuramoto phase-oscillator model produces locally correlated undulation but not a true head-to-tail traveling wave (Section 4.3). This is itself a result: the connectome topology, even with field coupling, does not generate the phase-lagged motor pattern that real locomotion requires. The traveling wave likely depends on circuit-level mechanisms (proprioceptive gain modulation, asymmetric gap junction rectification) that lie outside the current model class.

The kinematic scaffold isolates a specific question: *given the same locomotor capacity, do different coupling architectures produce different navigation performance through their effect on sensory-motor signal transmission?* The reversal decision reads AVA command interneuron activity from the neural dynamics. Sensory input (odor concentration) enters

through AWC/AWA neurons and must propagate through the connectome coupling to reach AVA. Different architectures provide different effective connectivity for this relay.

Results. Four architectures were tested with 20 matched runs each (4 initial headings $\times$ 5 random seeds, 120 s per run). All models show positive mean chemotaxis index, indicating successful navigation toward the food source (Figure 7a):

Architecture	CI (mean $\pm$ SEM)	Positive runs
Connectome only	+0.19 $\pm$ 0.11	14/20
+ Ephaptic	+0.21 $\pm$ 0.11	15/20
+ Neuropeptide	+0.24 $\pm$ 0.11	14/20
Trilayer	+0.26 $\pm$ 0.10	15/20

Because runs are matched by initial heading and seed, the paired comparison is appropriate. The trilayer architecture produces significantly higher CI than connectome-only (mean Δ CI = +0.071, paired t -test p = 0.015; Figure 7b). The neuropeptide-only improvement is marginal (Δ CI = +0.052, p = 0.073); the ephaptic-only improvement is not statistically significant (Δ CI = +0.018, p = 0.23). Reversal rates are slightly lower in the neuropeptide-containing models (2.15/min vs. 2.25/min; Figure 7c), consistent with improved sensory-motor transmission enabling more effective reversal suppression during up-gradient runs.

Interpretation. The result is narrow but real. Under identical locomotor conditions, richer coupling architectures improve navigation by improving the fidelity of sensory-to-motor signal transmission. The AWC $\rightarrow$ AIB $\rightarrow$ AVA relay that gates pirouette reversals involves neurons that are spatially clustered (body-axis positions 0.06–0.07), where ephaptic coupling is strongest, and connected by neuropeptide pathways that modulate effective gain. The trilayer provides parallel channels for this relay; the connectome alone provides only the sparse chemical synapses (AIB $\rightarrow$ AVA: 2–5 synapses, compared to thousands of total inputs to AVA).

This does not demonstrate that richer coupling produces better *locomotion*—the locomotor speed is prescribed. It demonstrates that the coupling architecture shapes the neural dynamics in ways that matter for behavior, even when locomotion is held constant. The distinction is important: the traveling-wave problem and the navigation problem are separable, and the current model addresses only the latter.

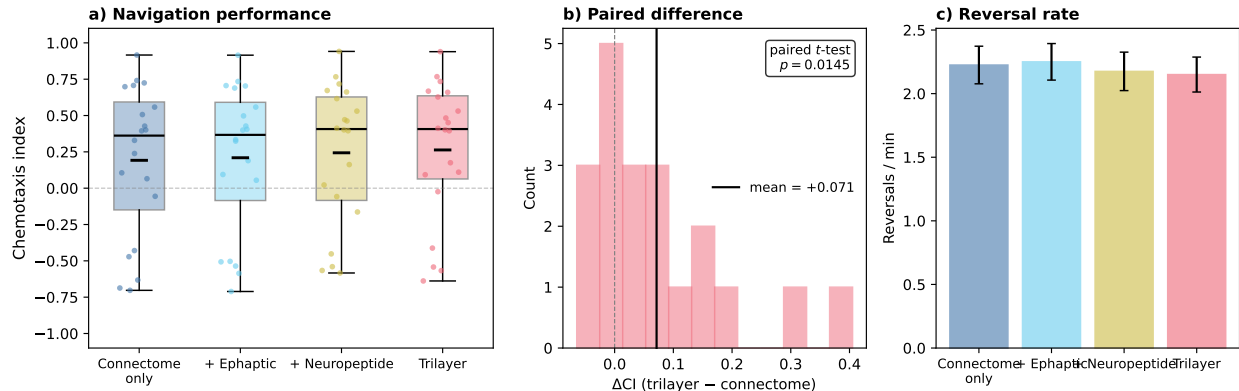


Figure 7: Kinematic chemotaxis comparison. Same 448-neuron connectome, four coupling architectures, 20 matched runs each (4 headings $\times$ 5 seeds, 120 s, kinematic speed 0.15 mm/s). **(a)** Chemotaxis index by architecture. All models show positive mean CI (navigation toward food). Boxes show quartiles; horizontal bars show means; points show individual runs. **(b)** Paired difference in CI between trilayer and connectome-only, matched by heading and seed. Mean shift = +0.071, paired t -test $p = 0.015$. **(c)** Mean reversal rate ($\pm$ SEM). Neuropeptide-containing models show slightly lower reversal rates, consistent with improved reversal suppression during up-gradient runs.

5 The Observational Gap

The *C. elegans* result illustrates a general principle: the connectome is a low-dimensional measurement of a high-dimensional system, and the measurement systematically underestimates what the system contains. This section demonstrates the phenomenon directly.

A coupled oscillator field ($25 \times 25 = 625$ oscillators, spatiotemporal OU environment, $\tau_{\text{env}} = 4$ s, coupling $K = 22$) is observed at varying resolution: from a single oscillator ($d_{\text{obs}} = 1$) to the full system ($d_{\text{obs}} = 625$). Observer dimensionality uses nested PCA truncation on the internal state $[\cos \theta_i, \sin \theta_i]$, guaranteeing monotonicity. The mutual information $I_{\text{pred}}(\Delta t)$ is computed at each resolution via Gaussian covariance estimation.

The result (Figure 8) is stark. The full system carries ~ 14 nats of predictive content. An observer with $d_{\text{obs}} = 5$ measures ~ 5 nats. The shaded region between them is *hidden correlation structure*: mutual information the system contains but the observer cannot verify. The gap is monotonic in observer resolution and opens once coupling is established—it is a property of the sync-measurement asymmetry, not of the specific system.

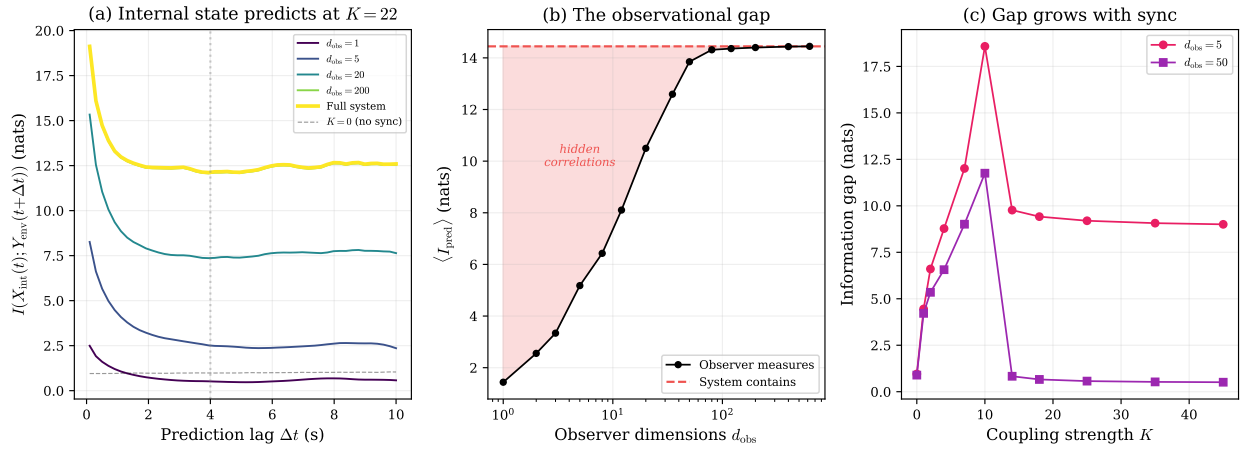


Figure 8: The observational gap. **(a)** Future MI at different observer resolutions. The full system (yellow) carries far more predictive information than any low-dimensional observer can verify. **(b)** Short-lag MI as a function of observer dimensionality. The red shaded region is hidden correlation structure. **(c)** The gap depends on coupling strength and observer resolution.

This has implications for how we account for thermodynamic order in biology. Synchronization creates distributed correlations across many coupled degrees of freedom that

coarse-grained observation necessarily destroys [15]. When the system later converts those hidden correlations into ordered behavior, the observer experiences this as order appearing without a corresponding information source. There is no second-law violation—the correlations were acquired through coupling, stored in the synchronized manifold, and spent at the behavioral output. But the observer’s entropy bookkeeping is incomplete because its measurement resolution is lower than the system’s internal dimensionality ($D_{\text{observer}} \ll D_{\text{eff}}$).

The connectome is exactly such a low-dimensional observer. It captures the sparse, directed connections that electron microscopy can resolve. It does not capture the continuous, spatially graded, temporally structured field dynamics that the coupling physics produces. The “mystery” of how 302 neurons generate complex adaptive behavior dissolves when the measurement instrument is recognized as lossy: the system has far more coupling channels than the connectome records.

In the spectral framework of Section 4.6, the observational gap has a concrete interpretation: it is the difference in spectral bandwidth between the organism’s full coupling architecture and the connectome alone. The hidden information—the shaded region in Figure 8—is precisely the medium and slow environmental modes that the ephaptic field and neuropeptide network track but the connectome cannot. A neuroscientist recording from *C. elegans* through calcium imaging and connectome analysis observes the fast synaptic dynamics but misses the slow neuropeptide landscape that determines behavioral state. The gap is not an artifact of measurement noise; it is a structural consequence of observing a multi-layer system through a single layer.

6 Prediction Without Coding

The predictive processing framework—predictive coding [40], the free-energy principle [41], active inference [42]—treats prediction as computation over internal generative models. We argue that this framework mistakes the shadow for the thing.

6.1 The category error

Coding is what happens when high-dimensional oscillatory dynamics collapse to a discrete output at the measurement boundary. It is the *product* of dimensional reduction, not the process that generates prediction. Predictive coding observes the low-dimensional residuals of high-dimensional synchronization and attributes them to an inferential process. But the prediction was already present in the synchronized manifold before any coding occurred.

Consider scope. Predictive coding requires a generative model, a hierarchical architecture, and an error-propagation mechanism. These exist in vertebrate cortex. They do not exist in bacteria, plants, cell collectives, or invertebrates with compact nervous systems—yet all exhibit anticipatory behavior. The synchronization framework provides a complete account at every scale: prediction arises from continuous coupling, not from inference.

6.2 What predictive coding actually describes

If prediction is synchronization, predictive coding describes the regime where synchronization *fails* and biology falls back on computation. This regime is real but rare. Computation becomes necessary in three conditions: (i) *adversarial environments*, where coupling channels carry systematically misleading structure; (ii) *absent coupling*, where the organism must predict in dimensions to which it is not currently coupled (counterfactual reasoning, planning in novel environments); (iii) *impoverished coupling*, where bandwidth is too narrow for relevant environmental structure. Most biological prediction—catching a ball, navigating a gradient, social synchrony in conversation—operates through continuous coupling where the synchronization framework provides a complete account without requiring generative models.

The free-energy principle’s central injunction—minimize surprise—can be restated as *maintain synchronization*: keep $\tau_{\text{sync}} < \tau_{\text{env}}$. The mathematics is the same; the interpretation differs. Under predictive coding, the organism *infers* environmental state through a model. Under the synchronization framework, the organism *is coupled to* the environment through

continuous dynamics. The first requires generative models. The second requires only a coupling surface. The first is phylogenetically recent. The second is as old as chemistry.

6.3 Coding as dimensional collapse

The oscillator field simulation demonstrates the cost of coding directly (Figure 9). The full continuous internal state of a 625-oscillator network carries ~ 14 nats of future MI with the environment. An explicit commit channel—projecting onto 5 dimensions, quantizing to 4 bits, updating at discrete intervals—retains only ~ 2 nats. Coding destroys $\sim 85\%$ of the predictive content. The lost information is precisely the distributed, high-dimensional correlation structure in phase relationships among coupled oscillators—the structure the synchronization framework identifies as the substrate of prediction.

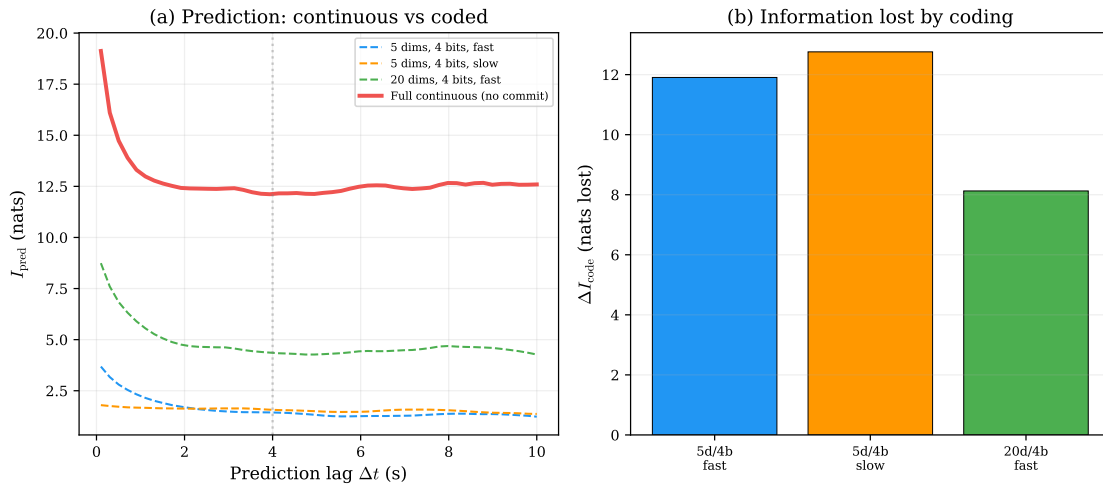


Figure 9: Prediction lives upstream of code. **(a)** Future MI for the full continuous state (red) versus explicit commit channels with different bandwidths (dashed). **(b)** Information lost by coding at short prediction lag.

Two decades of searching for the neural code of prediction have produced elegant theory and limited empirical confirmation. The synchronization framework suggests why: the neural activity that predictive coding interprets as “prediction signals” and “error signals” may be the low-dimensional shadow of high-dimensional synchronization dynamics. Looking for the

code is looking at commit events. The prediction lives in the oscillatory dynamics between commits.

7 Discussion

Simulation models. Three simulation approaches are presented. (1) *Fixed-coupling models* (Section 4.1): 448 coupled phase oscillators on the real connectome [4], with neuron positions from WormAtlas [5]. Three models (connectome only, +ephaptic, +full field) at three coupling regimes with five seeds each. $dt = 0.001\text{ s}$, $T = 50\text{ s}$, MI estimated via Gaussian covariance. (2) *Coherence field model* (Section 4.4): a Landau-Ginzburg order parameter $\varphi(x, t)$ on the body axis (200 grid points, $dt = 0.0002\text{ s}$, $T = 200\text{ s}$), bidirectionally coupled to the same 448-neuron connectome. (3) *Nonergodic defence model* (Sections 4.5–4.6): four model configurations (connectome only, +ephaptic, +neuropeptide, full trilayer) on a multi-mode oscillatory environment with 8 modes (timescales 0.1–1000 s, $1/f$ amplitude scaling) and metabolic budget constraint. $dt = 0.001\text{ s}$, $T = 1000\text{ s}$, coherence field on 100-point body-axis grid. The neuropeptide connectome (31,479 interactions) from Ripoll-Sánchez et al. [6] and 3D neuron positions from NeuroPAL [7] are integrated for the first time. The oscillator field simulation (Figures 1, 8, 9) uses a 25×25 lattice with nearest-neighbor coupling ($K_{\text{nn}} = 2$) driven by a spatiotemporal OU environment ($\tau_{\text{env}} = 4\text{ s}$, spatial correlation $L = 4$). Code is available at <https://github.com/todd866/predictive-horizons>.

7.1 Connection to the BioSystems program

This paper closes an arc across four prior contributions. *Limits of Falsifiability* [15] established that low-dimensional observers systematically underestimate high-dimensional systems—the connectome is exactly such an observer. *Timing Inaccessibility* [16] showed that measurement has a thermodynamic floor ($\tau_{\text{meas}} \geq k_{\text{B}}T \ln 2$ per bit)—this sets the bottleneck that synchronization outruns. *Intelligence as High-Dimensional Coherence* [17] argued that intel-

ligence is the maintenance of high-dimensional correlated structure—the nonergodic defence simulation demonstrates this concretely. *Coherence Time* [18] derived the physical timescale of oscillatory coordination (τ_{sync})—this determines which environmental modes each coupling layer can track.

The present paper combines these into a working model: the organism defends its nonergodic state by tracking environmental modes through multi-layer coupling, paying metabolic cost for each mode maintained. The distinction between information content and coordination capacity—the connectome carries information but cannot coordinate the body—extends to a deeper distinction: the connectome accesses only a narrow spectral band of the environment. The field layers extend the tracked bandwidth into the medium and slow modes that determine behavioral regime. Intelligence is not what the organism computes; it is how many environmental modes it can embody while remaining coherent.

7.2 Concentration of measure and the ridge of intelligence

The nonergodic defence result connects to a fundamental property of high-dimensional probability. Concentration of measure guarantees that in high dimensions, most probability mass lies on thin shells [15]. This is why high-dimensional projections look smooth and low-dimensional regardless of the underlying complexity. But it also means that a coherent high-dimensional system has robust statistical properties: its macroscopic behavior is predictable even when its microscopic state is unknowable. This is why intelligence requires high dimensionality—not because more dimensions mean more computation, but because concentration of measure provides free statistical regularity that low-dimensional systems must purchase through explicit control.

Intelligence lives on a ridge. Too few coherent dimensions: the system is rigid, cannot adapt, gets surprised by environmental changes it has no capacity to track. Too many dimensions without structure: the system is tracking noise, spending metabolic energy on modes with no predictive content. The width of this ridge is set by the environment’s

spectral complexity. A simple environment (few slow modes) rewards low D_{eff} . A complex environment (rich spectrum across timescales) rewards higher D_{eff} —up to the point where metabolic cost exceeds the predictive return. The trilayer architecture lets *C. elegans* operate on this ridge by providing coupling layers matched to different spectral bands, rather than forcing all prediction through a single narrow channel.

7.3 Progressive complexity loss and aging

Cohen et al. [43] documented progressive loss of physiological complexity with age: the degradation of high-dimensional oscillatory structure in cardiac, neural, and hormonal dynamics. The nonergodic defence framework provides a concrete mechanism. Aging is not the accumulation of damage to a static structure—it is the progressive contraction of the spectral bandwidth the organism can maintain.

As metabolic capacity declines, the organism can no longer sustain the neural synchrony that pumps φ across the full body axis. The neuropeptide network—the slowest, most metabolically expensive layer—degrades first, cutting access to the slow environmental modes. The system can still track fast signals through the connectome, but it loses the ability to anticipate slow regime changes. Each environmental mode lost is a prediction horizon shortened. The D_{eff} trajectory enters the “fragile” regime: perturbations that the young organism absorbs and recovers from now ratchet dimensionality downward. Eventually the system enters terminal decline—a self-reinforcing collapse in which lost modes reduce the capacity to detect and respond to the threats that further erode coherence.

This is falsifiable. The framework predicts that age-related cognitive decline should correlate with loss of slow oscillatory structure before loss of fast structure, and that metabolic interventions that restore slow-mode coupling should improve cognitive function even without repairing fast synaptic connections.

7.4 The substrate question

The coherence field model implies that biological intelligence requires a fundamentally different substrate than discrete computation. An oscillatory substrate maintains coherence natively—it re-entrains coupled elements every cycle, sustaining the synchronized manifold without explicit computation. A discrete substrate must *simulate* this coupling through sequential updates, paying an additional thermodynamic cost for each discrete approximation to continuous dynamics.

This is not an argument against artificial intelligence. It is an argument about efficiency. Oscillatory substrates defend nonergodicity cheaply because the physics of coupled oscillators provides coherence maintenance for free. Discrete substrates can achieve the same functional outcome, but at higher metabolic cost per mode tracked. The advantage of biological intelligence is not that it computes better, but that it computes less—coherence does the work that discrete systems must delegate to explicit processing.

7.5 Open questions

Coupling efficiency. Different biological architectures achieve different spectral bandwidths at different metabolic costs. Whether there exists a thermodynamic bound on coupling efficiency—a minimum cost per environmental mode tracked—is open. The framework of Still et al. [23] provides tools for formalizing it.

Evolutionary tuning. The framework predicts that selection on predictive capacity should drive expansion of the coupling interface and the number of coupling layers. The transition from bacterial chemoreception (one layer, fast modes only) through invertebrate multimodal sensing to vertebrate cortical integration (many layers, many timescales) can be read as an evolutionary trajectory toward broader spectral coverage.

Quantum anticipation. Igamberdiev’s program [37, 36] provides a complementary account grounded in quantum measurement theory. The present framework operates at the classical and mesoscopic level. Whether quantum constraints set floors on τ_{meas} more restrictive than

the classical Landauer bound is a question for Igamberdiev’s program; the timescale ordering $\tau_{\text{sync}} < \tau_{\text{meas}} < \tau_{\text{env}}$ would still organize the resulting dynamics.

7.6 Closing

The connectome is a remarkable scientific achievement—a complete wiring diagram of a nervous system. But a wiring diagram is a measurement, not a system. The connectome captures one coupling layer—fast, sparse, directed synaptic connections. The living worm uses at least three: the connectome for routing, the ephaptic field for spatial coordination, and the neuropeptide network for slow behavioral regime management. Each layer extends the spectral bandwidth of environmental modes the organism can track. Together, they let the worm defend its nonergodic state across the full range of ecological timescales.

The coherence domain—a spatially graded region of high φ , low-dimensional at macroscale, high-dimensional at microscale, maintained against noise by the neural synchrony pump—is a more physically precise object than the Markov blanket. Its gradients have measurable energetic cost ($\kappa|\nabla\varphi|^2$). Its persistence requires metabolic investment. Its contraction with metabolic decline provides a concrete, falsifiable account of aging as progressive loss of spectral bandwidth.

Intelligence is not computation. It is the thermodynamic work of maintaining a coherent, high-dimensional system on a structured manifold against the constant pressure of thermal noise—and the ecological work of ensuring that manifold tracks enough of the environment’s spectral structure to keep the organism alive. The field is right there. Evolution tunes it. Our simulations should model it.

Funding

No external funding supported this work.

Declaration of competing interest

The author declares no competing interests.

CRedit authorship contribution statement

Ian Todd: Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Software.

Acknowledgments

Connectome data from the OpenWorm project. Neuron positions from WormAtlas.

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